

Financial Time Series

Monday 4/26/21

Problem 29

data up to 4/1/21
generate 99% Conf interval for price
on 4/23/21 with no intervening data

Risk Mgt - future prediction

- (1) mean prediction (ARMA model)
 - (2) GARCH Volatility model GARCH(1,1)
current value at 4/1 - predict volatility
at 4/23
 - (3) Looking at stylized residuals (ARCH package)
Q-Q plot - dist of ϵ_t in log-return space
- ①

$$1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$X_t = \phi_0 + \phi_1 X_{t-1} + \varepsilon_t$$

$$\{ \varepsilon_t = 0, \dots, \varepsilon_{t-1}, \varepsilon_t \}$$

X_{t-1} is ε_{t-1} mean

But ε_t is indep of ε_{t-1}

$\Rightarrow X_{t-1}$ is indep of ε_t

$$a_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \omega + \alpha a_{t-1}^2$$

$$a_t^2 = \text{AR}(1) \text{ with } \phi_1 = \alpha$$

$$a_t^2 = \omega + \alpha a_{t-1}^2 + \eta_t$$

② $\{ \varepsilon_t, a_t \}$

$$1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\varepsilon_t = \sigma_t \varepsilon_t$$